

8. ADRs

ADR 1: Frontend framework

Context

To implement UX/UI and logic in web applications. We want to implement the project with ease, security and do not want to implement it with pure HTML.

Decision

Decided on the react framework. Typically, our team is familiar with react framework and it has many libraries including CSS and js library. As mentioned previously, we can implement the project rapidly and optimize time for development.

Status

Accepted

Consequences

Using react framework for frontend.

- The project is implemented rapidly.
- The many CSS frameworks are compatible.
- The react framework has a reuse component concept. We can decrease effort to create duplicate components.
- The document of react is quite poor.

ADR 2: CSS Framework

Context

To design layout and beautiful components in web applications. We want to implement layout and design of the web with ease and standard. And, we do not want to use pure CSS.

Decision

We will choose AntD for the CSS framework.

Status

Accepted

Consequences

Using AntD to implement layout and design of web applications.

- The web application has a beautiful user interface and is suitable for user experience.
- Some components are not in AntD. Therefore, we have to use another library.

ADR 3: Programming language

Context

We need to choose programming languages for our software. We have two major needs: a front-end programming language suitable for web applications, and a back-end programming language suitable for server applications

Decision

We will choose TypeScript for the front-end and Golang for the back-end.

Status

Accepted

Consequences

Using TypeScript for frontend and Golang for backend to implement web applications.

- TypeScript language helps our team and others to debug easier than JavaScript.
- TypeScript documents are quite complete and have a large community.
- Golang is the compiled language, it is obvious that performance is higher than an interpreted language like Javascript or Typescript. But, Golang is continually growing. It is important to be careful about versions and features that can be changed in major changes such as Golang 2.
- Golang is not pure OOP. We may be concerned about design patterns in some use cases.
- Some frameworks in applications may be deprecated.

ADR 4: Database for account management system, authentication, coin and payment system, and logging system

Context

The account management system, authentication, coin and payment system, and logging system have a specific structure. The schema of these systems will not frequently change, which means we can leverage the benefits of SQL databases to support these systems' design.

Decision

We will use MySQL to implement the above systems. There are 4 benefits to using MySQL as a database engine. These are data security and integrity, high performance, large community, and cost.

We will use the RDS of Amazon web service to deploy the MySQL database server, which can be operated 24/7, take a backup snapshot, and increase the database size automatically.

Status

Accepted

Consequence

Using MySQL leads to many of the following consequences.

- Development team must have at least one developer who can use Amazon web service and maintain database from RDS
- Developers must have knowledge and be familiar with SQL.

ADR 5: Database for user system, and video management system

Context

These two systems have an uncertain solution, which can be changed depending on business requirements. So, we study to find databases that can support rapid changes.

Decision

We will use MongoDB, which is a document database, to support these two systems' requirement

We will use MongoDB Atlas to deploy a MongoDB server, which can be operated 24/7, take a backup snapshot, and increase the database size automatically.

Status:

Accepted

Consequence

Using MongoDB leads to many of the following consequences.

- Development team must have at least one developer who can use MongoDB Atlas and maintain the database from it.
- Developers must have knowledge and be familiar with NoSQL and MongoDB syntax.

ADR 6: Environment Variable Configuration

Context:

The application should be configured by env file in order to change behavior depending on it such as different host, port number, and etc.

Decision:

We will use the .env file to configure the environment variable of the application.

We will use Viper library to load .env configuration for correct and fast operation.

Status:

Accepted

Consequence:

Using Viper to handle .env file format leads to many of the following consequences.

- If Viper's version is changed, testing is a must.
- If Viper will not support and update features further, we will change the library to handle the env file.

ADR 7: Figma for design UI

Context

The frontend team will implement UI following the UX/UI team. Therefore, we want to use tools for UI design and can work with our team in real-time.

UI designing has many tools for daft. But we want to use tools that are easy to use and can import CSS framework for daft prototype.

Decision

We will choose Figma for the design prototype.

Status

Accepted

Consequence

Using Figma for design prototype.

- It can import the AntD CSS framework for drag and drop components.
- The frontend team can work easily.
- It has many tools for design. However, some members are not familiar with it. Therefore, they have to take more time to learn.

ADR comments:

1. The decision for AntD was fine but it would be better if you also compare with some other CSS framework too.
2. In your ADR 3 about programming language, the bullets in your consequences about TypeScript / Golang should be moved to Context section. In the Consequences section, you should list what need to be prepared after the decision was made.
3. The consequence of your ADR 4 / ADR 5 however are fine.
4. I am still not sure why user system was decided to use MongoDB instead of a relational DB...
5. For the DB selections (ADRs 4 and 5), I also have another comment that you should also compare about the RDBMS such as MySQL / PostgreSQL. Or at least shows up that you are aware of some other available alternatives. This comment is also applied to ADR 6 about the selection of viper.
5. The consequence of ADR 7 about Figma, should be moved to Context as they explained why Figma is good or suit your need.